Effat University – College of Engineering – Department of Computer Science

CS4074: Big Data Analytics

Lab 2: Exploring Hadoop & HDFS

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Section: ______ **Date:** ______

Lab Overview

Objective:	Gain hands-on experience with Hadoop HDFS: file operations, block management, replication, fault tolerance, and storage design.
Prerequisites:	Hadoop installed and configured (Lab 1).
Duration:	2 hours (Parts 1–5: guided | Part 6: deep work | Part 7: deliverables).
Tools:	Terminal, Hadoop 3.x, Web browser (HDFS Web UI).

Lab Structure: Part 1: Start Services (10 min) → Part 2: HDFS Commands (20 min) → Part 3: Blocks & Replication (20 min) → Part 4: Admin & Health (15 min) → Part 5: Saudi Open Data Exercise (25 min) → Part 6: Deep Work (30 min) → Part 7: Deliverables (10 min)

Part 1: Verify Hadoop Installation & Start Services (10 min)

Before working with HDFS, we need to ensure Hadoop is properly running.

1.1 Step 1: Check Versions

Check Java & Hadoop

```
# Verify Java
java -version

# Verify Hadoop
hadoop version
```

Task 1.1

Run both commands and record the versions below:

- Java version: openjdk version "11.0.30" 2026-01-20
  OpenJDK Runtime Environment (build 11.0.30+7-post-Ubuntu-1ubuntu124.04)
  OpenJDK 64-Bit Server VM (build 11.0.30+7-post-Ubuntu-1ubuntu124.04, mixed mode, sharing)
- Hadoop version: Hadoop 3.3.6
  Source code repository https://github.com/apache/hadoop.git -r 1be78238728da9266a4f88195058f08fd012bf9c
  Compiled by ubuntu on 2023-06-18T08:22Z
  Compiled on platform linux-x86_64
  Compiled with protoc 3.7.1
  From source with checksum 5652179ad55f76cb287d9c633bb53bbd
  This command was run using /opt/hadoop/share/hadoop/common/hadoop-common-3.3.6.jar

1.2 Step 2: Start HDFS Services

Start HDFS

```
# Format NameNode (ONLY if first time -- do NOT repeat!)
# hdfs namenode -format

# Start HDFS daemons
start-dfs.sh

# Verify running processes
jps
```

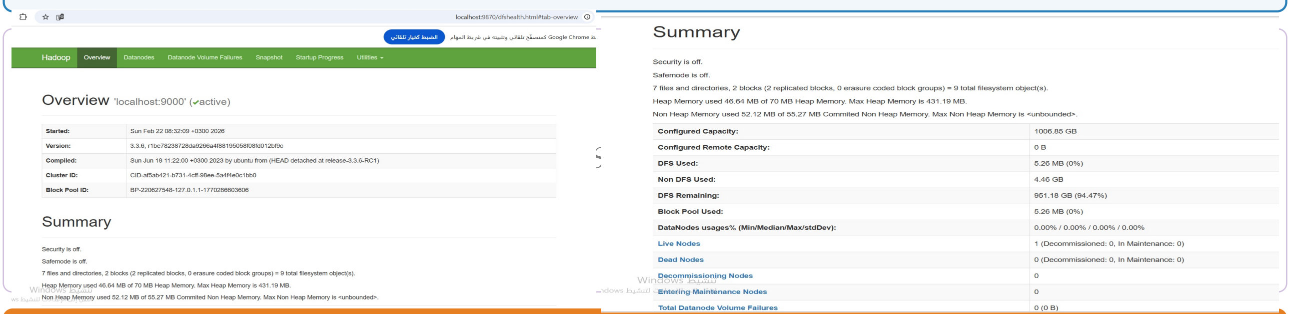
You should see at least: `NameNode`, `DataNode`, and `SecondaryNameNode`.

1.3 Step 3: Access the Web UI

Open your browser and navigate to: `http://localhost:9870`

Task 1.2

Take a screenshot of the HDFS Web UI overview page showing the cluster summary.



Common Issue

If `jps` does not show NameNode, try:

```
stop-dfs.sh
rm -rf /tmp/hadoop-*      # Clear temp data
hdfs namenode -format     # Re-format
start-dfs.sh
```

Part 2: Basic HDFS Commands (20 min)

HDFS commands follow the pattern: `hdfs dfs -<command> <arguments>`

2.1 Create a Working Directory

Directory Operations

```
# Create your lab directory in HDFS
hdfs dfs -mkdir /user
hdfs dfs -mkdir /user/student
hdfs dfs -mkdir -p /user/student/lab2/input
```

```
# List directories
hdfs dfs -ls /user/student/
hdfs dfs -ls -R /user/student/
```

2.2 Upload Files to HDFS

First, create a sample text file on your local machine:

Create Sample File

```
# Create a sample file locally
echo "Big Data Analytics is the future of computing.
Hadoop provides distributed storage with HDFS.
Apache Spark processes data 100x faster than MapReduce.
Saudi Vision 2030 embraces digital transformation.
Effat University leads in technology education." > ~/sample.txt

# Upload to HDFS
hdfs dfs -put ~/sample.txt /user/student/lab2/input/

# Alternative command (same result)
hdfs dfs -copyFromLocal ~/sample.txt /user/student/lab2/input/sample2.
    txt
```

2.3 Read & Browse Files

Reading Files in HDFS

```
# Display full file content
hdfs dfs -cat /user/student/lab2/input/sample.txt

# Show first 1KB
hdfs dfs -head /user/student/lab2/input/sample.txt

# Show last 1KB
hdfs dfs -tail /user/student/lab2/input/sample.txt

# Count lines, words, bytes
hdfs dfs -cat /user/student/lab2/input/sample.txt | wc -l
```

2.4 Download & Delete Files

Download & Cleanup

```
# Download from HDFS to local
hdfs dfs -get /user/student/lab2/input/sample.txt ~/downloaded.txt

# Copy within HDFS
hdfs dfs -cp /user/student/lab2/input/sample.txt /user/student/lab2/
    backup.txt

# Move / Rename
```

```
hdfs dfs -mv /user/student/lab2/backup.txt /user/student/lab2/renamed.
   txt

# Delete a file
hdfs dfs -rm /user/student/lab2/renamed.txt

# Delete a directory recursively
# hdfs dfs -rm -r /user/student/lab2/old_folder
```

Task 2.1

Complete the following exercises:

a) Create a directory: `/user/student/lab2/output`
b) Upload `sample.txt` into it
c) Display the contents using `-cat`
d) Count the number of words in the file
e) Download the file back to your local machine as `resu`

Write the exact commands you used:

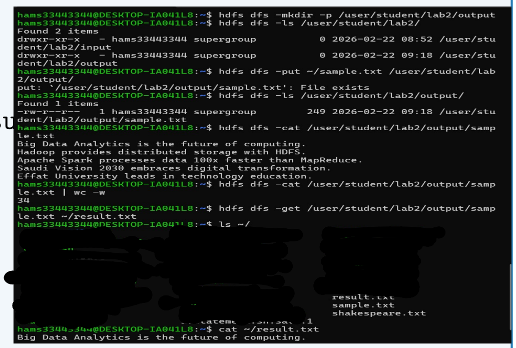
hdfs dfs -mkdir -p /user/student/lab2/output

hdfs dfs -put ~/sample.txt /user/student/lab2/output/

hdfs dfs -cat /user/student/lab2/output/sample.txt

hdfs dfs -cat /user/student/lab2/output/sample.txt | wc -w

hdfs dfs -get /user/student/lab2/output/sample.txt ~/result.txt



```
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -mkdir -p /user/student/lab2/output
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -ls /user/student/lab2/
Found 2 items
drwxr-xr-x   - hams33443344 supergroup          0 2026-02-22 08:52 /user/stu
dent/lab2/input
drwxr-xr-x   - hams33443344 supergroup          0 2026-02-22 09:18 /user/stu
dent/lab2/output
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -put ~/sample.txt /user/student/lab
2/output/
put: `/user/student/lab2/output/sample.txt': File exists
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -ls /user/student/lab2/output/
Found 1 items
-rw-r--r--   1 hams33443344 supergroup        249 2026-02-22 09:18 /user/stu
dent/lab2/output/sample.txt
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -cat /user/student/lab2/output/samp
le.txt
Big Data Analytics is the future of computing.
Hadoop provides distributed storage with HDFS.
Apache Spark processes data 100x faster than MapReduce.
Saudi Vision 2030 embraces digital transformation.
Effat University leads in technology education.
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -cat /user/student/lab2/output/samp
le.txt | wc -w
34
hams33443344@DESKTOP-IA041L8:~$ hdfs dfs -get /user/student/lab2/output/samp
le.txt ~/result.txt
hams33443344@DESKTOP-IA041L8:~$ ls ~/
[redacted]
                                                    sample.txt
                                                    shakespeare.txt
[redacted]
hams33443344@DESKTOP-IA041L8:~$ cat ~/result.txt
Big Data Analytics is the future of computing.
```

HDFS Command Reference

`-mkdir -p`	Create directory (with parents)
`-put / -copyFromLocal`	Upload local → HDFS
`-get / -copyToLocal`	Download HDFS → local
`-cat / -head / -tail`	Read file contents
`-ls / -ls -R`	List files (recursive)
`-cp / -mv`	Copy / Move within HDFS
`-rm / -rm -r`	Delete file / directory
`-du -h / -df -h`	Disk usage / free space
`-stat %r %b`	Show replication factor / block size
`-chmod / -chown`	Change permissions / ownership

Part 3: Understanding Blocks & Replication (20 min)

In Lecture 2 you learned that HDFS splits files into **128 MB blocks** and replicates each block **3 times**. Now let's see this in action.

3.1 Step 1: Generate a Large File

Generate a 300 MB File

```
# Generate a 300 MB file filled with random text
dd if=/dev/urandom of=~/bigfile.dat bs=1M count=300

# Verify size
```

```
ls -lh ~/bigfile.dat

# Upload to HDFS
hdfs dfs -put ~/bigfile.dat /user/student/lab2/input/
```

3.2 Step 2: Inspect Blocks

Check Block Distribution

```
# See how HDFS split the file into blocks
hdfs fsck /user/student/lab2/input/bigfile.dat -files -blocks

# Check replication factor
hdfs dfs -stat %r /user/student/lab2/input/bigfile.dat

# Check block size
hdfs dfs -stat %b /user/student/lab2/input/bigfile.dat
```

Task 3.1

Record the output from `hdfs fsck`:

- Total file size: 314572800 bytes (300 MB)
- Number of blocks: 3 blocks
- Replication factor: 1
- Total replicated blocks: 3x1=3

Reflection 3.1

A 300 MB file with 128 MB block size should produce how many blocks? Show your calculation. Does this match your `fsck` output?

300 ÷ 128 = 2.34
HDFS rounds up to the next whole number.
So:
2 full blocks + 1 partial block = 3 blocks
This matches the fsck output which shows 3 blocks.

3.3 Step 3: Change Replication Factor

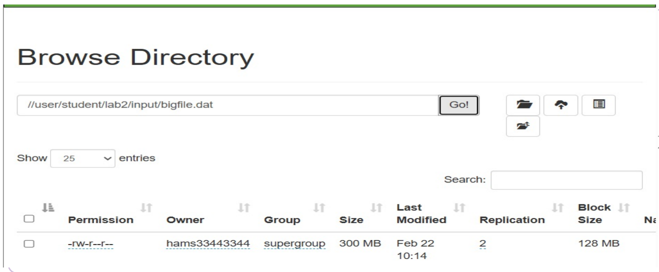
Modify Replication

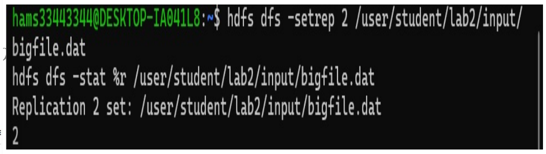
```
# Change replication factor to 2
hdfs dfs -setrep 2 /user/student/lab2/input/bigfile.dat

# Verify the change
hdfs dfs -stat %r /user/student/lab2/input/bigfile.dat

# Check under-replicated blocks (if any)
hdfs fsck /user/student/lab2/input/bigfile.dat -files -blocks
```

Task 3.2

a) What was the replication factor before? 1
b) What is it now? 2

c) Open the Web UI (`http://localhost:9870`) → Utilities → Browse the file system. Take a screenshot showing the file and its replication.





Part 4: HDFS Admin & Health Check (15 min)

As a system administrator, you need to monitor HDFS health.

4.1 Cluster Report

HDFS Admin Report

```
# Full cluster report
hdfs dfsadmin -report
```

Task 4.1

From the `dfsadmin -report` output, fill in:

- Total configured capacity: 1006.85 GB
- DFS used: 1015.00 MB
- DFS remaining: 949.86 GB
- Number of live DataNodes: 1

4.2 Disk Usage

Disk Usage Commands

```
# Show disk usage of your files (human readable)
hdfs dfs -du -h /user/student/lab2/

# Show total HDFS capacity
hdfs dfs -df -h
```

4.3 Safe Mode

Safe Mode Operations

```
# Check if in safe mode
hdfs dfsadmin -safemode get

# Enter safe mode (read-only -- no writes allowed)
hdfs dfsadmin -safemode enter
```

```
# Try uploading -- this should FAIL!
echo "test" > /tmp/test.txt
hdfs dfs -put /tmp/test.txt /user/student/lab2/

# Leave safe mode
hdfs dfsadmin -safemode leave

# Now the upload should work
hdfs dfs -put /tmp/test.txt /user/student/lab2/
```

Task 4.2

a) Enter safe mode and attempt to upload a file. What error message do you get?

When attempting to upload a file in safe mode, the system returned the following error:
"Cannot create file /user/student/lab2/test.txt.*COPYING*. Name node is in safe mode."

b) Why would an administrator use safe mode? Give one real-world scenario.

Safe mode prevents any write operations to HDFS. An administrator uses safe mode during system startup, maintenance, or recovery. For example, after restarting the NameNode, safe mode ensures that all data blocks are properly replicated and verified before allowing new data to be written to the system.

4.4 File System Check

HDFS Health Check

```
# Check entire HDFS for corrupt/missing blocks
hdfs fsck / -files -blocks

# Summary only
hdfs fsck /
```

Reflection 4.1

What does `hdfs fsck` report about the health status? Are there any corrupt or missing blocks?

The hdfs fsck command reports that the filesystem status is HEALTHY. There are no corrupt or missing blocks. However, there are 3 under-replicated blocks because the replication factor was set to 2 while only one DataNode is available. Despite this, the system is functioning correctly and no data loss has occurred.

Part 5: Real-World Exercise: Saudi Open Data (25 min)

In this part, you will work with real data and organize it in HDFS like a professional data engineer.

5.1 Step 1: Prepare the Data

Generate a Realistic Dataset

```
# Create a Saudi cities population dataset (CSV)
cat > ~/saudi_population.csv << 'EOF'
city,region,population_2024,area_km2,density
Riyadh,Riyadh,7500000,1798,4171
Jeddah,Makkah,4700000,1686,2788
Makkah,Makkah,2200000,760,2895
```

```
Madinah,Madinah,1500000,589,2547
Dammam,Eastern,1300000,800,1625
Khobar,Eastern,650000,750,867
Tabuk,Tabuk,600000,780,769
Abha,Asir,500000,370,1351
Taif,Makkah,700000,321,2181
Buraidah,Qassim,600000,980,612
EOF

# Verify
cat ~/saudi_population.csv
wc -l ~/saudi_population.csv
```

5.2 Step 2: Create Organized HDFS Structure

Organize Data in HDFS

```
# Create a professional directory structure
hdfs dfs -mkdir -p /data/saudi/population
hdfs dfs -mkdir -p /data/saudi/weather
hdfs dfs -mkdir -p /data/saudi/healthcare

# Upload the dataset
hdfs dfs -put ~/saudi_population.csv /data/saudi/population/

# Verify the structure
hdfs dfs -ls -R /data/saudi/

# View file in HDFS
hdfs dfs -cat /data/saudi/population/saudi_population.csv
```

5.3 Step 3: Analyze with Basic Commands

Basic Data Analysis via CLI

```
# Count total records (minus header)
hdfs dfs -cat /data/saudi/population/saudi_population.csv | tail -n +2 |
    wc -l

# Find cities in Makkah region
hdfs dfs -cat /data/saudi/population/saudi_population.csv | grep "Makkah
   "

# Sort by population (descending)
hdfs dfs -cat /data/saudi/population/saudi_population.csv | tail -n +2 |
    \
    sort -t',' -k3 -nr

# Show only city names and populations
hdfs dfs -cat /data/saudi/population/saudi_population.csv | tail -n +2 |
    \
    cut -d',' -f1,3
```

Task 5.1

Using HDFS commands, answer the following:

a) How many cities are in the dataset? 10

b) Which city has the highest population? Riyadh,Riyadh,7500000,1798,4171

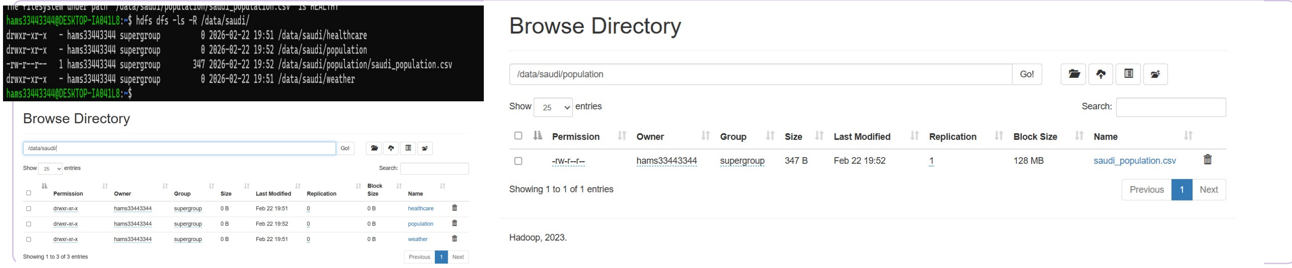
c) How many cities are in the Makkah region? 3

d) What is the block count for this CSV file? (`hdfs fsck`) 1 block.

e) Why is the CSV stored in only 1 block? The CSV file is stored in only one block because its size is much smaller than the default HDFS block size (128 MB). Since the file is very small, it does not need to be split into multiple blocks.

Task 5.2

Take a screenshot of the HDFS Web UI showing the `/data/saudi/` directory tree.



Part 6: Deep Work – Fault Tolerance & Storage Experimentation (30 min)

About Deep Work

This section challenges you to **investigate HDFS behavior beyond the basics**. You will design experiments, observe system behavior, and draw conclusions that connect to Lecture 2 concepts. Deep work questions require **critical thinking** — there are no single right answers, but your reasoning must be supported by experimental evidence.

6.1 Experiment 6.1: The Small Files Problem

In Lecture 2, we discussed that HDFS struggles with many small files. Let's **prove it**.

Small Files Experiment

```
# Step 1: Record baseline NameNode status
hdfs dfsadmin -report | grep -E "files|blocks|Total"

# Step 2: Generate 1,000 small files (1 KB each)
mkdir -p ~/small_files
for i in $(seq 1 1000); do
    dd if=/dev/urandom of=~/small_files/file_$i.txt bs=1K count=1 2>/dev
      /null
done

# Verify: total size is ~1 MB
du -sh ~/small_files/

# Step 3: Upload all small files to HDFS
hdfs dfs -mkdir -p /user/student/lab2/small_files
hdfs dfs -put ~/small_files/* /user/student/lab2/small_files/
```

```
# Step 4: Check NameNode status again
hdfs dfsadmin -report | grep -E "files|blocks|Total"

# Step 5: Check total blocks used
hdfs fsck /user/student/lab2/small_files/ | grep -E "Total blocks|Total
    size"
```

Now let's compare with an archive approach:

Archive the Small Files

```
# Step 6: Create a Hadoop Archive (HAR)
hadoop archive -archiveName small_files.har -p /user/student/lab2/
    small_files \
     /user/student/lab2/

# Step 7: List the archive contents
hdfs dfs -ls har:///user/student/lab2/small_files.har/

# Step 8: Check blocks used by the archive
hdfs fsck /user/student/lab2/small_files.har/ -files -blocks | \
     grep -E "Total blocks|Total size"
```

Task 6.1 – Record Your Findings

Metric	1,000 Small Files	HAR Archive
Total data size	374,784 B (~ 366 KB)	409,809 B (~400 KB)
Number of HDFS blocks	366	3
NameNode entries (files + blocks)	366 file entries + 366 blocks entires	1 archive file + 3 blocks

Reflection 6.1

Why does 1 MB of data stored as 1,000 files consume significantly more NameNode resources than a single 1 GB file? Explain using the concept that each file/block entry takes ∼150 bytes of NameNode memory.

Although 1 MB of total data is very small compared to 1 GB, storing it as 1,000 separate files creates 1,000 file metadata entries and 1,000 block entries in the NameNode. Each file and each block requires approximately 150 bytes of memory in the NameNode. Therefore, 1,000 small files may consume around 300,000 bytes (1000 files + 1000 blocks × 150 bytes) of NameNode memory. In contrast, a single 1 GB file might only consist of a few large blocks (for example, 8 blocks if using 128 MB block size), resulting in far fewer metadata entries. This demonstrates that NameNode memory usage depends on the number of files and blocks, not the total data size, which explains why many small files consume significantly more NameNode resources.

6.2 Experiment 6.2: Simulating Node Failure

Let's test HDFS fault tolerance by manually corrupting a block.

Fault Tolerance Experiment

```
# Step 1: Upload a test file
echo "This is a critical healthcare record." > ~/critical.txt
hdfs dfs -put ~/critical.txt /user/student/lab2/critical.txt
```

```
# Step 2: Find block locations
hdfs fsck /user/student/lab2/critical.txt -files -blocks -locations

# Step 3: Find the actual block file on the local filesystem
# (Look for the block ID from Step 2)
find /tmp/hadoop-*/dfs/data -name "blk_*" -type f 2>/dev/null | head -10

# Step 4: Corrupt a block by overwriting it
# COPY the block ID from Step 2, then:
# find /tmp/hadoop-*/dfs/data -name "blk_<BLOCK_ID>*" -type f
# echo "CORRUPTED" > <path_to_block_file>

# Step 5: Wait ~30 seconds, then check
sleep 30
hdfs fsck /user/student/lab2/critical.txt -files -blocks

# Step 6: Can you still read the file?
hdfs dfs -cat /user/student/lab2/critical.txt
```

Task 6.2 – Record Your Observations

a) How many replicas did the file have before corruption? The file had a replication factor of 1 (one replica).

b) After corrupting one block, could you still read the file? Yes, the file was still readable using the `hdfs dfs -cat` command.

c) Did HDFS detect the corruption? How? In this experiment, no corruption was detected because the block file was not successfully modified. The FSCK report showed the file status as HEALTHY with zero corrupt blocks.

d) Did HDFS re-replicate the block to restore the replication factor? No re-replication occurred because the replication factor was 1 and no corruption was successfully introduced.

Reflection 6.2

How long did recovery take? What would happen if **2 out of 3** replicas were lost simultaneously? At what point does HDFS lose data permanently?

In this experiment, the replication factor was set to 1, meaning only one copy of the block existed. Since the block was not successfully corrupted, recovery was not required. If the replication factor had been 3 and one replica was lost, HDFS would automatically re-replicate the block from the remaining healthy copies. However, if 2 out of 3 replicas were lost, the system could still recover using the remaining replica. HDFS permanently loses data only when all replicas of a block are lost. Recovery time depends on cluster size and network speed, but in small clusters it typically takes a few seconds.

6.3 Experiment 6.3: Block Size Experiment

Varying Block Size

```
# Upload the SAME 300 MB file with different block sizes

# Block size = 64 MB
hdfs dfs -D dfs.blocksize=67108864 -put ~/bigfile.dat \
    /user/student/lab2/bigfile_64m.dat

# Block size = 128 MB (default)
hdfs dfs -D dfs.blocksize=134217728 -put ~/bigfile.dat \
    /user/student/lab2/bigfile_128m.dat

# Block size = 256 MB
hdfs dfs -D dfs.blocksize=268435456 -put ~/bigfile.dat \
    /user/student/lab2/bigfile_256m.dat

# Compare block counts
```

```
echo "=== 64 MB blocks ==="
hdfs fsck /user/student/lab2/bigfile_64m.dat 2>/dev/null | grep "Total
   blocks"

echo "=== 128 MB blocks ==="
hdfs fsck /user/student/lab2/bigfile_128m.dat 2>/dev/null | grep "Total
   blocks"

echo "=== 256 MB blocks ==="
hdfs fsck /user/student/lab2/bigfile_256m.dat 2>/dev/null | grep "Total
   blocks"
```

Task 6.3 – Fill the Table

File Size	Block Size	Number of Blocks	NameNode Entries	
300 MB	64 MB	5	5	
300 MB	128 MB	3	3	
300 MB	256 MB	2	2	

Reflection 6.3

When would you choose a **smaller** block size vs. a **larger** one? Discuss the trade-offs in terms of: (1) NameNode memory, (2) parallelism in MapReduce, and (3) network overhead.

Smaller block sizes increase the number of blocks, which increases NameNode memory usage because each block requires metadata storage. However, smaller blocks improve parallelism in MapReduce jobs because more tasks can run simultaneously. Larger block sizes reduce the number of metadata entries, saving NameNode memory, but decrease parallel processing opportunities. Larger blocks also reduce network overhead because fewer blocks need to be transferred. Therefore, smaller blocks are useful for high parallel processing workloads, while larger blocks are better for reducing metadata and network overhead.

6.4 Experiment 6.4: Build a Data Lake Directory Structure

Design and implement an HDFS directory structure for a hospital system.

Data Lake Design

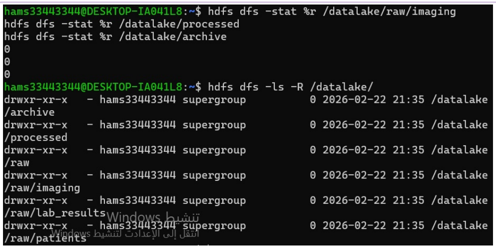
```
# Create the hospital data lake structure
hdfs dfs -mkdir -p /datalake/raw/patients
hdfs dfs -mkdir -p /datalake/raw/lab_results
hdfs dfs -mkdir -p /datalake/raw/imaging
hdfs dfs -mkdir -p /datalake/processed
hdfs dfs -mkdir -p /datalake/archive

# Set different replication factors
hdfs dfs -setrep 3 /datalake/raw/imaging
hdfs dfs -setrep 2 /datalake/processed
hdfs dfs -setrep 1 /datalake/archive

# Verify replication settings
echo "--- Imaging replication ---"
hdfs dfs -stat %r /datalake/raw/imaging
echo "--- Processed replication ---"
hdfs dfs -stat %r /datalake/processed
```

```
echo "--- Archive replication ---"
hdfs dfs -stat %r /datalake/archive

# View the full tree
hdfs dfs -ls -R /datalake/
```

Task 6.4

a) Take a screenshot of `hdfs dfs -ls -R /datalake/` output.

b) Upload a sample file to each of: `/datalake/raw/patients/`, `/datalake/raw/imaging/`, and `/datalake/archive/`. Verify their replication factors differ using `-stat %r`.



lake directory structure screenshot here

The hospital data lake directory structure was successfully created in HDFS under `/datalake/`. The structure includes raw data (patients, lab results, imaging), processed data, and archive directories. Replication factors were configured according to data importance. The directory tree was verified using the `hdfs dfs -ls -R /datalake/` command.

Reflection 6.4

Why might a hospital want **different replication factors** for X-ray images (rep=3) vs. old archived logs (rep=1)? Consider: data criticality, storage cost, and recovery requirements.

A hospital may assign different replication factors based on data criticality. X-ray images are highly critical medical records and must remain available even if a node fails, so a replication factor of 3 ensures high fault tolerance. Processed data requires moderate reliability, so replication factor 2 is sufficient. Archived logs are less critical and accessed infrequently, so replication factor 1 reduces storage costs while still maintaining basic availability. This strategy balances data protection, storage cost, and recovery requirements.

Part 7: Deliverables & Submission **(10 min)**

Submission Checklist

	Item	Description	Marks
	Screenshot 1	HDFS Web UI overview (Task 1.2)	1
	Task 2.1	HDFS commands for file operations	2
	Task 3.1	Block inspection results from `fsck`	2
	Task 3.2	Replication factor change + screenshot	2
	Task 4.1	Admin report values	1
	Task 4.2	Safe mode experiment + answers	2
	Task 5.1	Saudi data analysis answers	2
	Screenshot 5.2	HDFS directory structure	1
	Task 6.1	Small files experiment table + reflection	3
	Task 6.2	Fault tolerance observations + reflection	3
	Task 6.3	Block size table + reflection	3
	Task 6.4	Data lake structure + screenshot + reflection	3
		Total	**25**

Submission Instructions:

- Submit this completed lab document as a **PDF** with all screenshots, answers, and reflections.
- File name format: `CS4074_Lab2_YourName_YourID.pdf`
- Deadline: **One week from today's lab session.**
- Late submissions: **−5 marks per day.**

Before You Leave

```
# Stop HDFS when done
stop-dfs.sh

# Verify all daemons are stopped
jps
```

End of Lab 2

Next Lab: MapReduce Programming